Extracting DNA from Fruit

Biology 2e chapter 14 reading connection | Teacher guide / classroom preparation

Learning target

How does the extraction mixture affect visible material recovered from fruit?

Timing and materials

One class period; chill alcohol and pilot extraction beforehand.

Strawberries, sealable bags, water, salt, plain dish detergent, filters and funnels, clear tubes, measuring cylinders, cold school-approved ethanol or isopropyl alcohol, goggles and wooden sticks.

Classroom route

1. Teacher pilot starting recipe: 90 mL water, 10 mL plain detergent and 1 g salt. Prepare a matched mixture replacing detergent with water. Use 10 g fruit and 10 mL mixture per trial; the blank has mixture but no fruit.

2. Mash each fruit portion consistently, mix gently with its assigned solution, and filter. Transfer 5 mL filtrate to a tube and gently layer 5 mL cold alcohol. Use identical alcohol type and concentration throughout. Keep alcohol away from flames and heat; do not taste materials.

3. Wait five minutes, observe without disturbing layers, and score strands using one shared rubric. Repeat each treatment three times. Dispose according to school rules. Do not interpret crude precipitate as pure DNA.

Pilot the physical procedure with your own equipment. Follow school handling and disposal rules and the purchased kit instructions. The virtual numerical settings are not an equipment prescription.

Extracting DNA from Fruit

Biology 2e chapter 14 reading connection | Teacher guide / instruction and assessment

Hybrid or substitute assignment

Assign all virtual, all physical, or specified samples in each mode. The classroom entry button is available during collection. Students complete the same prediction and evidence questions. Record classroom time series on the worksheet; the online classroom form stores endpoint measurements and notes. Keep sources and measurement scales separate.

Guided virtual sequence

Students prepare each sample, inspect the apparatus or classify specimens, then explicitly collect data. All assigned trials are required before exploration. Reload pauses the clock and restores saved work. Downloads retain source labels; no automatic grade is submitted.

Expected reasoning

The model complete extraction usually has more visible strands than its no-detergent control; the blank has none. Real precipitate is not pure DNA. Visibility scores are ordered categories, not concentrations; compare distributions and descriptions rather than treating scores as mass.

Feedback and scoring

Criterion	Points	Evidence
Prediction and controls	0-2	Testable claim; comparable samples and controlled variables.
Observations	0-2	Complete, labeled records with units and source.
Interpretation	0-2	Correct calculations or categorical comparisons; claim tied to evidence.
Limitations	0-2	Specific uncertainty and a useful follow-up investigation.

Score reasoning and data quality, not agreement of physical results with the simulation. Manual tallies are student observations; a matching total does not prove every category is correct. Do not replace student counts with the model answer. The optional Sordaria extension retains classification feedback.

Extracting DNA from Fruit

Biology 2e chapter 14 reading connection | Teacher guide / model and sources

Model limitations

Illustrative visibility, not molecular replication or quantitative DNA yield. Fruit chemistry and detergent formulation vary. This activity supports DNA isolation; replication needs a separate evidence investigation.

Troubleshooting

Confirm the preparation steps are finished before starting or collecting. Corn and fly activities require manual tallies summing to the sample size plus review; optional specimen marks do not classify or tally automatically. Chromosome models require all four gametes plus a reviewed tally; cell measurements require a focused 40x view; gels require staining and three size estimates; fluorescence labs require the viewer. Use the visible Collect data and continue button to advance.

For physical work, preserve unexpected observations, check the control and instrument setup, then repeat with fresh material where appropriate. Never replace measured results with the virtual answer.

Evidence and optional classroom resources

DNA extraction background

https://www.genome.gov/about-genomics/teaching-tools/strawberry-dna-extraction

External resources remain on their publishers' websites. This packet contains original text and tables, not supplier worksheets. Chapter numbers are reading connections. No supplier or textbook publisher endorsement is implied.

Teacher pilot checklist

Confirm availability and preparation time; test controls; check units and observation windows; verify handling/disposal; choose class grouping and accessible alternatives; keep notes for revision.